

## TROYES-BARBEREY AP, France

WMO: 071680

Lat: 48.3247N

Lon: 4.0200E

Elev: 118

StdP: 99.91

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -6.9       | -4.5 | -10.3                         | 1.6 | -6.0 | -7.5 | 2.0 | -3.3 | 14.0                  | 9.8  | 12.5 | 9.8  | 2.2                   | 300  | 0.549 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 12.6                   | 32.0            | 20.4 | 29.9 | 19.8 | 27.9 | 19.3 | 21.6                | 29.5 | 20.8 | 28.1 | 19.8 | 26.3 | 3.6                  | 80   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 19.1                            | 14.1 | 24.4 | 18.2 | 13.3 | 23.3 | 17.5 | 12.7 | 22.4 | 63.5          | 29.7 | 60.3 | 28.3 | 57.1 | 26.3 | 25.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 10.9              | 9.4  | 8.2 | -10.1                      | 36.0 | 2.6                | 2.5 | -11.9                                              | 37.8 | -13.4      | 39.2 | -14.9      | 40.6 | -16.7      | 42.4 |
|                   |      |     | -10.5                      | 23.6 | 2.6                | 0.9 | -12.3                                              | 24.2 | -13.9      | 24.8 | -15.3      | 25.3 | -17.2      | 25.9 |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|      |                                                                   | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|-------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                   | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 11.1  | 3.6   | 4.3   | 7.1   | 10.0  | 13.8  | 17.2  | 19.6  | 19.1  | 15.4  | 11.9  | 7.0   | 4.3  | (6)  |
| (7)  |                                                                   | DBStd       | 6.70  | 4.33  | 3.95  | 3.44  | 3.30  | 3.31  | 3.35  | 3.20  | 3.29  | 2.99  | 3.71  | 3.87  | 4.13 | (7)  |
| (8)  |                                                                   | HDD10.0     | 816   | 201   | 160   | 101   | 39    | 6     | 0     | 0     | 0     | 1     | 23    | 103   | 181  | (8)  |
| (9)  |                                                                   | HDD18.3     | 2779  | 458   | 392   | 347   | 250   | 146   | 60    | 23    | 29    | 96    | 202   | 339   | 437  | (9)  |
| (10) |                                                                   | CDD10.0     | 1236  | 2     | 2     | 12    | 39    | 123   | 217   | 298   | 282   | 163   | 82    | 14    | 3    | (10) |
| (11) |                                                                   | CDD18.3     | 155   | 0     | 0     | 0     | 0     | 4     | 27    | 62    | 52    | 9     | 1     | 0     | 0    | (11) |
| (12) |                                                                   | CDH23.3     | 2015  | 0     | 0     | 0     | 14    | 85    | 370   | 759   | 646   | 129   | 11    | 0     | 0    | (12) |
| (13) |                                                                   | CDH26.7     | 704   | 0     | 0     | 0     | 1     | 12    | 116   | 284   | 261   | 30    | 0     | 0     | 0    | (13) |
| (14) | Wind                                                              | WSAvg       | 3.7   | 4.2   | 4.2   | 4.0   | 3.6   | 3.4   | 3.2   | 3.4   | 3.2   | 3.3   | 3.5   | 3.8   | 4.2  | (14) |
| (15) | Precipitation                                                     | PrecAvg     | 677   | 49    | 48    | 45    | 52    | 62    | 54    | 58    | 61    | 53    | 70    | 61    | 66   | (15) |
| (16) |                                                                   | PrecMax     | 938   | 115   | 96    | 116   | 129   | 142   | 105   | 144   | 122   | 135   | 152   | 148   | 162  | (16) |
| (17) |                                                                   | PrecMin     | 513   | 9     | 6     | 7     | 1     | 17    | 12    | 7     | 9     | 10    | 24    | 17    | 10   | (17) |
| (18) |                                                                   | PrecStd     | 103   | 23    | 25    | 26    | 31    | 27    | 23    | 32    | 34    | 31    | 35    | 31    | 33   | (18) |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB    | 13.7  | 16.0  | 20.9  | 25.2  | 28.2  | 33.3  | 35.2  | 36.8  | 29.7  | 24.9  | 18.2  | 14.4 | (19) |
| (20) |                                                                   |             | MCWB  | 11.6  | 10.6  | 12.6  | 16.5  | 19.6  | 21.6  | 21.3  | 21.2  | 18.9  | 17.9  | 14.5  | 11.9 | (20) |
| (21) |                                                                   | 2%          | DB    | 12.3  | 13.2  | 18.1  | 22.4  | 25.6  | 29.8  | 31.9  | 32.1  | 27.0  | 22.0  | 16.0  | 12.9 | (21) |
| (22) |                                                                   |             | MCWB  | 10.6  | 9.8   | 11.7  | 14.8  | 18.0  | 20.5  | 20.2  | 20.4  | 18.6  | 16.8  | 13.5  | 11.1 | (22) |
| (23) |                                                                   | 5%          | DB    | 11.0  | 11.6  | 15.5  | 19.7  | 23.6  | 27.4  | 29.8  | 29.6  | 24.2  | 19.8  | 14.3  | 11.6 | (23) |
| (24) |                                                                   |             | MCWB  | 9.4   | 9.2   | 10.8  | 13.4  | 17.0  | 19.6  | 19.7  | 19.7  | 17.4  | 15.9  | 12.4  | 10.1 | (24) |
| (25) |                                                                   | 10%         | DB    | 9.6   | 10.2  | 13.4  | 17.1  | 21.4  | 25.2  | 27.8  | 27.0  | 22.1  | 17.7  | 12.9  | 10.3 | (25) |
| (26) |                                                                   |             | MCWB  | 8.3   | 8.4   | 10.0  | 12.0  | 15.7  | 18.4  | 19.1  | 18.7  | 16.2  | 14.7  | 11.3  | 9.0  | (26) |
| (27) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB    | 12.2  | 11.7  | 14.1  | 17.6  | 20.9  | 22.9  | 22.5  | 22.7  | 20.5  | 18.5  | 15.5  | 12.7 | (27) |
| (28) |                                                                   |             | MCDWB | 13.3  | 14.5  | 18.5  | 23.8  | 26.7  | 30.0  | 31.3  | 31.5  | 26.6  | 23.0  | 17.4  | 13.8 | (28) |
| (29) |                                                                   | 2%          | WB    | 10.9  | 10.5  | 12.5  | 15.4  | 18.9  | 21.4  | 21.4  | 21.5  | 19.1  | 17.4  | 13.7  | 11.3 | (29) |
| (30) |                                                                   |             | MCDWB | 12.1  | 12.5  | 16.4  | 21.1  | 24.2  | 28.0  | 29.7  | 29.9  | 25.1  | 21.1  | 15.5  | 12.6 | (30) |
| (31) |                                                                   | 5%          | WB    | 9.6   | 9.5   | 11.5  | 13.8  | 17.5  | 20.2  | 20.5  | 20.4  | 18.1  | 16.2  | 12.6  | 10.2 | (31) |
| (32) |                                                                   |             | MCDWB | 10.8  | 11.2  | 14.7  | 18.5  | 22.4  | 26.2  | 28.1  | 27.8  | 23.2  | 19.1  | 14.2  | 11.5 | (32) |
| (33) |                                                                   | 10%         | WB    | 8.3   | 8.5   | 10.4  | 12.5  | 16.2  | 19.0  | 19.6  | 19.4  | 17.0  | 15.1  | 11.4  | 9.1  | (33) |
| (34) |                                                                   |             | MCDWB | 9.5   | 10.1  | 13.1  | 16.3  | 20.3  | 24.2  | 26.2  | 25.4  | 21.2  | 17.4  | 12.8  | 10.2 | (34) |
| (35) | Mean Daily Temperature Range                                      | 5% DB       | MDBR  | 5.9   | 7.6   | 9.9   | 11.5  | 11.5  | 12.3  | 12.6  | 12.5  | 11.6  | 9.6   | 6.8   | 5.8  | (35) |
| (36) |                                                                   |             | MCDBR | 6.9   | 10.1  | 14.4  | 16.5  | 15.7  | 16.6  | 17.0  | 17.7  | 15.8  | 12.4  | 8.8   | 7.1  | (36) |
| (37) |                                                                   |             | MCWBR | 5.7   | 7.3   | 8.9   | 9.7   | 9.1   | 8.3   | 7.0   | 7.4   | 8.2   | 7.8   | 6.5   | 5.6  | (37) |
| (38) |                                                                   | 5% WB       | MCDBR | 6.8   | 8.8   | 12.0  | 14.5  | 13.9  | 14.7  | 14.8  | 15.1  | 13.4  | 11.1  | 8.3   | 6.4  | (38) |
| (39) |                                                                   |             | MCWBR | 5.8   | 6.6   | 7.9   | 8.9   | 8.5   | 7.9   | 6.4   | 7.1   | 7.3   | 7.2   | 6.4   | 5.4  | (39) |
| (40) | Clear-Sky Solar Irradiance                                        | taub        | 0.317 | 0.340 | 0.369 | 0.399 | 0.406 | 0.414 | 0.410 | 0.401 | 0.372 | 0.362 | 0.331 | 0.314 | (40) |      |
| (41) |                                                                   | taud        | 2.431 | 2.376 | 2.301 | 2.221 | 2.246 | 2.260 | 2.263 | 2.309 | 2.369 | 2.394 | 2.429 | 2.404 | (41) |      |
| (42) |                                                                   | Ebn at Noon | 750   | 811   | 841   | 849   | 856   | 850   | 847   | 838   | 829   | 772   | 723   | 698   | (42) |      |
| (43) |                                                                   | Edh at Noon | 66    | 86    | 109   | 130   | 132   | 131   | 129   | 118   | 101   | 84    | 66    | 60    | (43) |      |
| (44) | All-Sky Solar Radiation                                           | RadAvg      | 0.93  | 1.72  | 2.94  | 4.37  | 5.15  | 5.80  | 5.51  | 4.78  | 3.69  | 2.21  | 1.11  | 0.76  | (44) |      |
| (45) |                                                                   | RadStd      | 0.13  | 0.27  | 0.38  | 0.53  | 0.56  | 0.53  | 0.48  | 0.48  | 0.33  | 0.16  | 0.11  | 0.14  | (45) |      |

## Historical Trends

|      |                         | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|-------|---------|-------|-----|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|      |                         | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j) | (k)         | (l)     | (m)     |         |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A     | N/A     |         |
| (47) | Regional (35 neighbors) | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A     | N/A     |         |

Nomenclature: See separate page